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Synthetic Network Traffic Generation for Intrusion Detection Systems: a Systematic Literature Review

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Abstract. Network data can be difficult to collect due to privacy and confidentiality reasons. For these reasons, network datasets are typically created with controlled environments called testbeds. However, these datasets are regularly criticized for their limited size, class imbalance, obsolescence, and lack of actual user activity. Following the rapid development of generative artificial intelligence, new methods have been applied to synthetic network traffic generation without emulation or simulation. This systematic literature review assesses the current state of synthetic network traffic generation for intrusion detection systems.

Keywords: Networking · Generative AI · Traffic Generation

1 Introduction

Data is central in many applications: to analyze and visualize the state of a system, to train AI-based models, to assess processes, etc. However, accessing such data in computer networks can be challenging due to their sensitive and private nature. For this reason, network analysis generally relies on controlled and simplified environments, called testbeds, that do not contain private or sensitive information. However, testbeds have limitations too. First, they lack real user behavior and suffer from a limited diversity of technologies and usages. Due to this issue, it may not be possible to extrapolate results obtained on a testbed, such as an intrusion detection system, to a network with actual, more complex and varied user behavior. Second, collecting data is slow: to generate a dataset including several weeks of data, such as CICIDS2018 [57], the testbeds must run for several weeks. So, when errors [31] happen during an experiment, rerunning can be very costly. Besides, it also means that it is difficult to quickly create a dataset that contains recent technologies or attacks, contributing to the overall obsolescence of datasets.

Synthetic network traffic generators create network traffic without emulation or simulation. Molnar et al. [41] identified five categories of synthetic network traffic generators: replay engines, maximum throughput generators, model-based generators, high-level and auto-configurable generators, and special scenario generators. Due to the recent surge of generative artificial intelligence, this study

is focused on model-based generators. New models such as Generative Adversarial Networks (GAN), Variational Auto-Encoder (VAE), diffusion models, and Large Language Models (LLMs) are more and more used in academia to generate synthetic network traffic. These models are generally trained on network traffic datasets and can generate synthetic data far faster than a testbed. However, their quality may not be up to par with that of testbeds.

This article proposes a Systematic Literature Review (SLR) [26] on this topic.

2 Background and Scope of the Study

Writing a survey involves summarizing many works to produce and analyze a snapshot of scientific effort within some scope. However, many biases can alter such surveys. Important articles may be missed, such as work from other scientific communities than the authors, using different keywords, or published in less prestigious venues. Systematic Literature Review (SLR) [26] is a methodology that notably aims at reducing such biases with a systematic and reproducible procedure. This procedure can be summarized in the following steps: 1) defining the scope of the study; 2) identifying precise research questions; 3) systematically collecting articles by using regular expressions on specialized search platforms; 4) reading and analyzing them.

The subject of synthetic network traffic generation is at the intersection of three domains: networking, cybersecurity, and machine learning. To deal with the large variety of works, we reduce the scope of the study with three restrictions. First, we limit our study to IT networks based on the IP protocol, because most of the literature on intrusion detection systems and synthetic network traffic generation focuses on such networks. In particular, we do not include works on industrial (OT) networks, IoT networks, or 5G networks unless they are generic enough to be applied to IT networks. Second, as motivated in the introduction, we only consider *model-based* generation, i.e., generation relying on machine learning or statistical methods. In particular, a large body of work focuses on hand-written generation rules. Such rules are generally designed for a specific need. They are not scalable and cannot accommodate the large variety of attacks or user behavior in IT networks. Third, we focus on applications for cybersecurity, such as intrusion detection systems, honeynets for malware analysis, or environments for cybersecurity experts' training.

3 Research Questions Definition

We propose the following research questions as a way to build a snapshot of the practices of the synthetic network traffic generation community:

RQ1: *Which communities work on synthetic network traffic generation?*

RQ2: *What are the applications of synthetic network traffic generation in cybersecurity?*

RQ3: *What types of data are generated?*

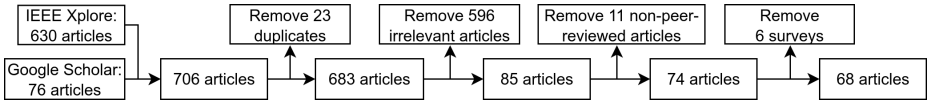

Fig. 1. The PRISMA diagram of this systematic literature review.

RQ4: *What generation techniques are used?*
RQ5: *How are generated data evaluated?*
RQ6: *What open-source implementations exist?*
RQ7: *What are the performances of the generators?*

4 Corpus Creation

We based our systematic literature enumeration on the IEEE Xplore and Google Scholar search engines. IEEE Xplore supports regular expressions for searching. After reviewing several works in the domain, we decided on this expression:

```
(("Document Title":"generative" OR "Document Title":"synthetic"
 OR "Document Title":"generation" OR "Document Title":"generator"
 OR "Document Title":"data augmentation")
 AND ("Document Title":"network data"
 OR "Document Title":"network traces" OR "Document Title":"traffic"
 OR "Document Title":"packet" OR "Document Title":"pcap"
 OR "Document Title":"log" OR "Document Title":"internet")
 AND NOT "Document Title":"next generation")
```

We search Google Scholar with the following keywords: `generative synthetic generation "data augmentation" network traffic packet pcap internet`.

The article collection was performed on April 28th and updated on August 8th. We collected 630 articles published since 2010 from IEEE Xplore. Google Scholar does not provide a definite list of relevant articles, so we collected articles until the results were no longer relevant. The complete selection process is detailed in Fig. 1. The final corpus is composed of 68 articles. We did not filter the articles by the rank of their conference nor by the number of citations.

5 Answers to Research Questions

5.1 RQ1: Which communities work on synthetic network traffic generation?

Synthetic network traffic generation is a multidisciplinary topic. In our corpus, 37 articles have been published in networking-adjacent venues, 10 in cybersecurity-adjacent venues, 4 in AI-adjacent venues, and 17 in other venues. Details are available in the Appendix. Generally, AI models are mere tools and cybersecurity is only considered a possible application. Surprisingly, the cybersecurity community seems to downplay the importance and the impact of synthetic data.

5.2 RQ2: What Are the Application of Synthetic Network Traffic Generation in Cybersecurity?

We identified two main applications with different challenges. Some articles do not clearly state their applications and have been classified to our best guess.

Data Augmentation (30 articles) The motivation for data augmentation is the imbalance in network datasets, e.g., if there is more benign traffic than malicious traffic. This imbalance typically causes performance issues with machine learning algorithms [64]. Data augmentation proposes to generate more examples of the minority classes (generally attacks) to reduce this imbalance. This application aims to expand real data with synthetic ones, not to supplant them.

Dataset Generation (38 articles) Dataset generation can be useful when actual data is scarce or difficult to share due to privacy or confidentiality issues. Such datasets can be used for machine learning training or classifier evaluation. For some works, the goal is to simulate benign, real-world traffic, e.g., for digital twins [73] or cybersecurity exercises [28]. Articles may focus on generating just one data class, such as video traffic [25] or a specific attack [42].

Data augmentation is more common in cybersecurity (50% of these articles use data augmentation) to enhance detection performances and less prevalent in the networking community (only 32%) that focuses more on building and analyzing datasets.

5.3 RQ3: What Types of Data Are Generated?

There are four main types of generated data. Table 1 details which data type is more common for different applications. One method can generate several types of data.

Traffic Statistics (3 articles) Traffic statistics describe the traffic without detailing each flow e.g., with throughput time series [44, 49, 25]. This is the least common data type.

Flow Statistics (41 articles) Flow statistics are the most commonly generated data, notably for data augmentation. Indeed, many IDS rely on flows for their decision. Since there is no standard flow statistics, there is a wide variety of output formats: while some articles generate 5-uplets, others create up to 84 statistics, which is in line with what the CICFlowMeter tool produces.

Packet Metadata (25 articles) Due to the widespread use of cryptography in packet payloads, packet metadata is considered sufficient for many applications. The nature of these metadata depends from one article to another: e.g., [56] generates a sequence of packet sizes, and [47] generates sequences of TCP flags, inter-packet arrival times, and packet sizes.

Packet Payloads (16 articles) The last category of generated data is packet payloads. Packet payloads are necessary to send packets on the wire, but are rarely analyzed by security tools due to privacy concerns and the widespread use of cryptography.

The most common format for data augmentation is flow statistics. Indeed, flow statistics are tabular data, making them very suitable for a wide range of machine learning techniques. For dataset generation, while flow statistics are also popular, a large proportion generates pcap files with packet payload and metadata, making them applicable to a wider range of usages.

Applications	Data augmentation	Dataset generation
Traffic	0	3
Flow	**23**	**18**
Packet metadata only	3	8
Packet payload only	1	1
Packet metadata and payload	3	11

Table 1. Number of articles per types of generated data and applications.

5.4 RQ4: What Generation Techniques Are Used?

Various techniques are used in model-based generation. The most popular techniques are deep learning techniques [32], namely Generative Adversarial Networks (GAN), Variational AutoEncoder (VAE), diffusion models and Large Language Models (LLMs). Surprisingly, *all but one article* [71] of the corpus experiment with a deep learning method, either in a new method or as a baseline.

Generative Adversarial Networks (47 articles) Vanilla GAN are by far the most used techniques. The most common variants are CGAN, which allow for conditional generation (11 articles), CTGAN, which allow for conditional generation and are specialized in tabular data (10 articles), and WGAN (13 articles), a more classical flavor of GAN. A few articles use CNN-GAN to generate images ([72, 11, 9, 68]) and a few architectures are used to generate time sequences (RCGAN, TimeGan, and SigCWGAN). Besides, some articles introduce their own GAN variants, such as CCGAN [51], PcapGAN [18] or MF-CGAN [50]. The reader is encouraged to read this survey [45] on GAN in networking to learn more about these models. Overall, GAN are present in 47 articles, i.e., 69% of the corpus.

Other Deep Learning Techniques (24 articles) The second most popular technique is VAE (and variations), with 8 articles: [14, 56, 67, 1, 47, 54, 62, 40]. Diffusion and LLMs are more recent techniques. Diffusion is used in 7 articles ([22, 67, 23, 73, 24, 77, 60]) and LLMs in 5 articles ([39, 27, 15, 36, 16]). A few more deep learning techniques are used, such as Seq2Seq (an ancestor of transformers) by [34], LSTM by [43], transformers by [59, 76] and convolutional networks by [20, 70]. Figure 2 presents the number of times a deep learning technique is used per year. First, we can remark that the popularity of the field is accelerating quickly.

Second, the proportion of articles using GAN is decreasing and new techniques, such as diffusion and LLMs, are gaining popularity.

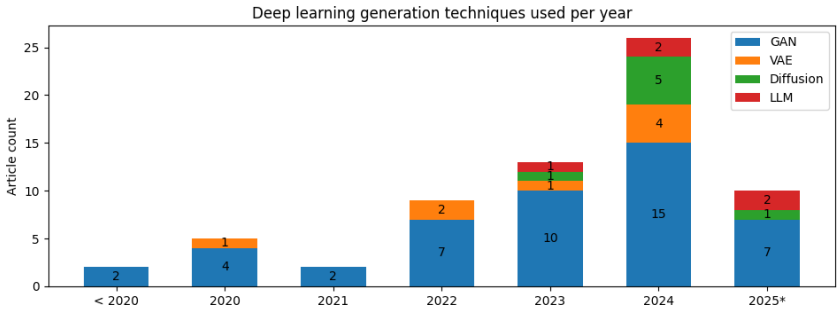

Fig. 2. The number of occurrences of deep learning generation techniques across the years. The year 2025 is incomplete.

Other Techniques (8 articles) Some other techniques have been used, such as SMOTE [10] (by [14, 67, 43, 30, 39, 65]) that was specifically introduced for data augmentation, its variant ADASYN (by [67]), Bayesian networks (by [55, 15]), a graphical and probabilistic model, and network flow pattern mining (by [15]).

5.5 RQ5: How Are Generated Data Evaluated?

A wide variety of metrics have been proposed to evaluate synthetic data.

TSTR (47 articles) The most common way in our corpus to evaluate synthetic data is called TSTR (for "Train on Synthetic, Test on Real"), proposed in this domain by [80]. This metric does not assess the data themselves but their effect on a downstream task, typically intrusion detection performance.

Marginal distribution distance (19 articles) Such metrics compare the distribution of each features independently. The metrics include Earth Mover Distance (8 articles: [56, 47, 72, 55, 15, 21, 24, 53]) for continuous features, and Jensen-Shannon Divergence (9 articles: [22, 70, 1, 72, 55, 15, 24, 53, 77]) or Kullback-Leibler Divergence (2 articles: [44, 53]) for categorical features.

Qualitative visual evaluation (16 articles) Such metrics typically rely on dimension reduction techniques such as PCA or t-SNE.

Similarity distance (15 articles) This metrics compare data points. This metrics include cosine similarity ([58, 54, 21, 34, 78], the Euclidian distance ([54, 21]), Total Variation Distance ([22, 33, 77]), the Hellinger distance ([22]), the distance to nearest neighbors ([8]), the Fréchet Inception Distance between images ([7, 60]) and the L_1 metric ([42]).

Networking metrics (11 articles) These metrics are specific to the networking domain. For example, "packet generation success rate" evaluates the proportion of packet that can be correctly sent to the network ([27, 11, 48, 38, 7, 16]) and Domain Knowledge Check (DKC) verifies some network rules (port 53 should use UDP, ICMP packets should have empty payloads, etc.) in the generated data ([70, 13, 55, 15, 16]).

Pairwise correlations (5 articles) These metrics compare pairwise features. A few metrics have been used, such as pairwise correlation ([55, 15, 8]), Pearson correlation ([54, 33]) and Spearman correlation ([54]).

The two applications use different metrics. The TSTR metric have become standard for data augmentation: 97% of the article using data augmentation rely on the TSTR metric. However, such a metric does not easily apply to dataset generation, as one seeks to assess data quality itself and not necessarily through its utility. A few articles focus on evaluation metrics, such as [67] and [55], and they highlight the requirements of using multiple metrics to assess the quality according to different criteria (such as diversity, realism and fidelity) and different scale (packets-level or flow-level).

5.6 RQ6: What open-source implementations exist?

Out of the 68 articles we examined, only 14 referenced an open-source implementation, and only 5 were replicated (details are available in the Appendix). Some work may simply be too recent, published the same year as this systematic literature review. Overall, the vast majority of the articles do not compare empirically their contribution to the state of the art. In fact, except NetShare, these methods were only replicated once. This could be caused by the diversity of generated data (flow statistics or pcap, attack or benign) and a lack of standard baselines accepted in the community.

5.7 RQ7: What are the performances of the generators?

It is difficult to conclude on the effectiveness of the vast majority of the articles of our corpus because they generate different kinds of data, for different purposes, evaluate with different metrics, rely on different datasets and do not compare their results to baselines. Nonetheless, there is an overall consensus from the articles that do compare their methods to baselines.

Impact of data augmentation for classification In [14], SMOTE [10], a classical oversampling method, outperforms CTGAN, Copula GAN, and VAE in a data augmentation task. The systematic evaluation of [67] revealed that most resampling strategies negatively impacted the classification performance across all models. [39] has a similar conclusion: the best classifiers used no data augmentation or SMOTE. In [30], SMOTE is second to WGANGP. In [43], SMOTE is better than CNN-based generation, and in [65], the method proposed by the authors is better than random oversampling (ROS), which is better than GAN.

Dataset generation quality In [55], Bayesian networks outperform CTGAN, E-WGAN-GP, and NetShare, and NetShare is worse than a method that samples each feature independently. In [15], a mix of Bayesian networks and data mining outperforms TVAE, CTGAN, E-WGAN-GP, a method based on GPT2, and NetShare on the same metrics as [55].

Overall, the classical SMOTE algorithm can be very effective and should be systematically used as a baseline. Even if deep learning methods are commonly used, they are consistently outperformed by simpler methods. This conclusion is consistent with findings in other communities, such as image classification [63].

6 Conclusion

Model-based synthetic network traffic generation is an active research domain with an increasing number of articles that benefit from the current surge of generative AI. However, the performance of deep learning methods over statistical methods has not yet been proven and the utility of these methods is still discussed. We believe synthetic network traffic generation could be more helpful to the cybersecurity community. To that end, we propose a few open challenges to the community.

Challenge 1 Define a standard benchmark with a set of metrics that could capture the quality of synthetic network traffic, a standard network dataset, and standard baselines. The set of metrics should include temporal consistency and non-functional characteristics of the generator (such as training and generation times). The standard baselines should include state-of-the-art methods and simple methods like SMOTE and random oversampling (ROS) as a low baseline and real data, as proposed in [55], as a high baseline. Such a benchmark could help articles compare each other.

Challenge 2 Concept drift (the fact that the behavior of a network changes over time) is an operational issue. Research on the subject is difficult due to the lack of appropriate datasets. Thus, cybersecurity could benefit from datasets that include concept drift [6]. However, this subject is never mentioned in our corpus.

Challenge 3 Recent cybersecurity datasets include APT-like behavior to challenge IDS with stealthy, multi-step, and long-time attacks. The articles of our corpus that create synthetic attacks for dataset generation are limited to single-step attacks. Testbed-based generation cannot easily create months-long datasets and tends to perform APT attacks within a week, which is unrealistic. Synthetic datasets could fill that gap since the generation time is generally much shorter than the dataset duration.

The surge of deep learning methods for synthetic network traffic generation is still not backed by performance increases. Nonetheless, synthetic network traffic generation could become more useful in cybersecurity by focusing more on dataset generation that could foster new research directions.

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Appendix

This appendix includes the subset of corpus associated to several research questions.

RQ1 The articles published in networking-adjacent venues are: [1, 9, 11, 13, 15, 22, 23, 25, 29, 5, 21, 33, 38, 40, 43, 46–50, 53, 61, 68, 69, 71, 36, 72, 75, 7, 3, 79, 74, 16, 76, 65, 60, 77]. The articles published in cybersecurity-adjacent venues are: [35, 37, 54–56, 62, 66, 67, 70, 78]. The articles published in AI-adjacent venues are: [18, 20, 24, 51]. The articles published in other venues are: [14, 27, 34, 39, 44, 52, 58, 73, 4, 17, 30, 59, 8, 19, 12, 42, 28].

RQ2 The articles performing data augmentation: [22, 67, 13, 39, 51, 50, 18, 75, 35, 9, 43, 46, 34, 68, 66, 37, 3, 79, 4, 17, 30, 59, 8, 78, 19, 12, 76, 74, 20, 58]. The articles performing dataset generation: [14, 11, 69, 25, 55, 15, 61, 36, 5, 27, 72, 62, 38, 49, 53, 42, 1, 65, 77, 60, 16, 7, 28, 33, 71, 21, 73, 48, 47, 70, 56, 54, 52, 23, 29, 24, 40, 44].

RQ3 The articles generating flow statistics: [14, 70, 67, 5, 13, 54, 52, 39, 50, 72, 69, 29, 75, 55, 15, 35, 21, 43, 73, 24, 33, 46, 34, 20, 66, 53, 40, 3, 79, 4, 17, 30, 8, 78, 19, 12, 42, 76, 77, 68, 74]. The articles generating packet metadata: [22, 58, 28, 47, 23, 27, 72, 18, 71, 62, 11, 9, 33, 61, 48, 38, 37, 40, 7, 59, 16, 60, 65, 1, 56]. The articles generating packet payload: [51, 22, 23, 27, 18, 71, 62, 11, 9, 33, 48, 38, 36, 7, 16, 65].

RQ4 The articles using GAN: [14, 58, 56, 67, 28, 5, 13, 54, 52, 51, 50, 72, 18, 11, 69, 25, 29, 75, 55, 35, 21, 9, 33, 48, 46, 38, 49, 68, 66, 37, 53, 7, 30, 78, 19, 12, 76, 74, 44]. The articles using CGAN: [58, 5, 13, 50, 21, 49, 66, 67, 3, 30, 65]. The articles using CTGAN: [14, 54, 72, 75, 55, 53, 4, 17, 30, 78]. The articles using WGAN: [56, 13, 72, 25, 55, 7, 79, 30, 8, 78, 42, 74, 61]. Some articles experiment with several variants of GAN.

RQ5 The articles using TSTR: [14, 22, 58, 70, 56, 67, 1, 47, 13, 54, 39, 51, 50, 23, 72, 18, 25, 75, 15, 35, 9, 43, 73, 24, 61, 46, 44, 20, 68, 66, 37, 53, 3, 79, 4, 17, 30, 59, 8, 78, 19, 12, 42, 76, 60, 65, 74, 44]. The articles using qualitative visual evaluation: [58, 47, 50, 18, 69, 21, 73, 24, 38, 44, 49, 40, 3, 17, 59, 8].

RQ6 Our findings are summarized in Table 2.

Method	Year	GitHub Repository	Replicated?
SIP-GAN [38]	2021	amarmeddahi/sip-gan	No
STAN [70]	2021	ShengzheXu/stan	Yes (e.g., [1])
NetShare [72]	2022	netsharecmu/NetShare	Yes (e.g., [55])
NeCSTGen [40]	2022	fmeslet/NeCSTGen	Yes (e.g., [62])
SyNIG [61]	2023	Nirhoshan/SyNIG	No
PAC-GPT [27]	2023	dark-0ne/NetworkPacketGenerator	No
NetDiffusion[22]	2024	noise-lab/NetDiffusion_Generator	No
FlowChronicle [15]	2024	joschac/FlowChronicleCoNEXT	No
CGAN-based-Tool [5]	2024	Alanoud-Subahi/CGAN-based-Tool	No
NetDiffus [60]	2024	Nirhoshan/NetDiffus	Yes (e.g., [77])
GAN-based [78]	2024	ydataai/ydata-synthetic	Yes (e.g., [2])
PNetGPT [36]	2025	Snail1502/PNetGPT	No
GAN-based [19]	2025	i-am-junayed/XAI-Intrusion-Detection-System	No
GPT on the wire [16]	2025	javieradelgado/GPT-on-the-wire	No

Table 2. Number of articles per types of generated data and applications. To access the repository, prefix them with `https://github.com/`. Replication is to the best of our knowledge at time of writing.